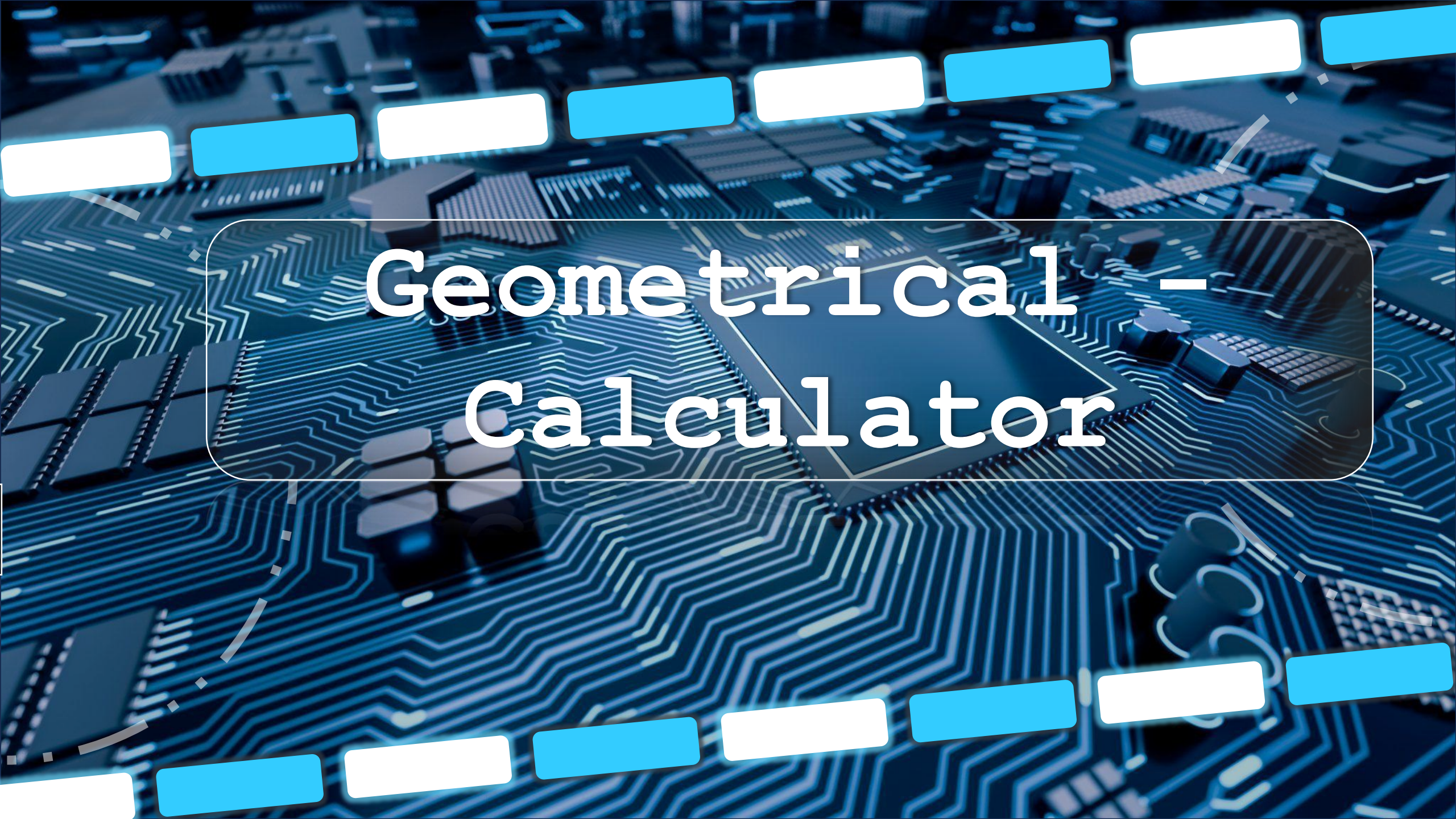
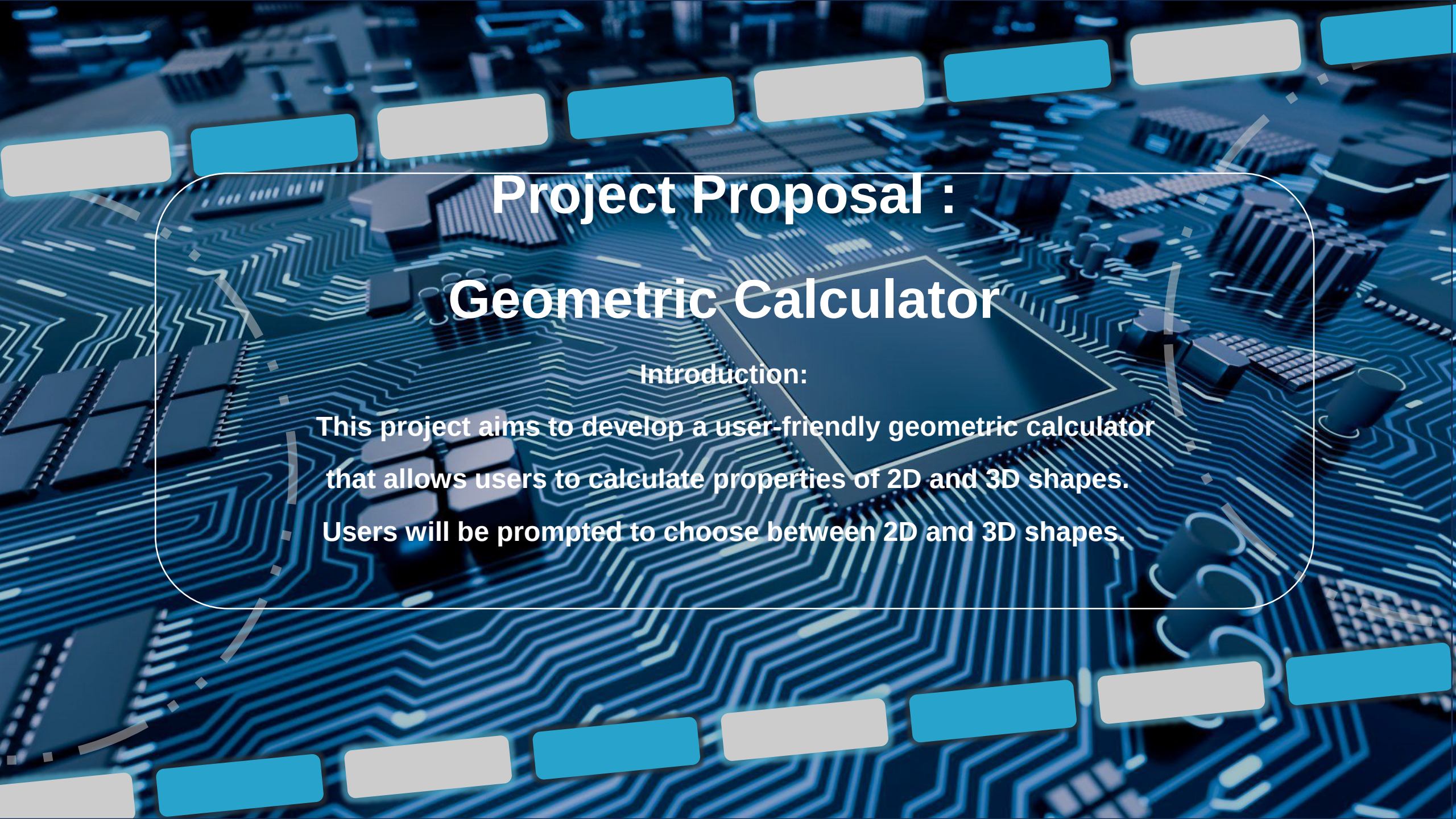




Geometrical – Calculator



Project Proposal : Geometric Calculator

Introduction:

This project aims to develop a user-friendly geometric calculator that allows users to calculate properties of 2D and 3D shapes. Users will be prompted to choose between 2D and 3D shapes.

Objective

Menu System

Develop a user-friendly menu system for users to choose between 2D and 3D shapes.

2D Shapes

Implement functions for calculating the area and circumference of 2D shapes (e.g., Circle, Rectangle).

3D Shapes

Implement functions for calculating the surface area and volume of 3D shapes (e.g. Sphere).

User Experience

Provide a seamless user experience with clear prompts and options.

Working

1. Selection

Users choose between 2D and 3D shapes in the main menu.

2. 2D Shapes 2

Sub-menu options for calculations (e.g., area, circumference) based on shape selection.

3. Input

Users input the required parameters for the chosen calculation.

4. Calculation

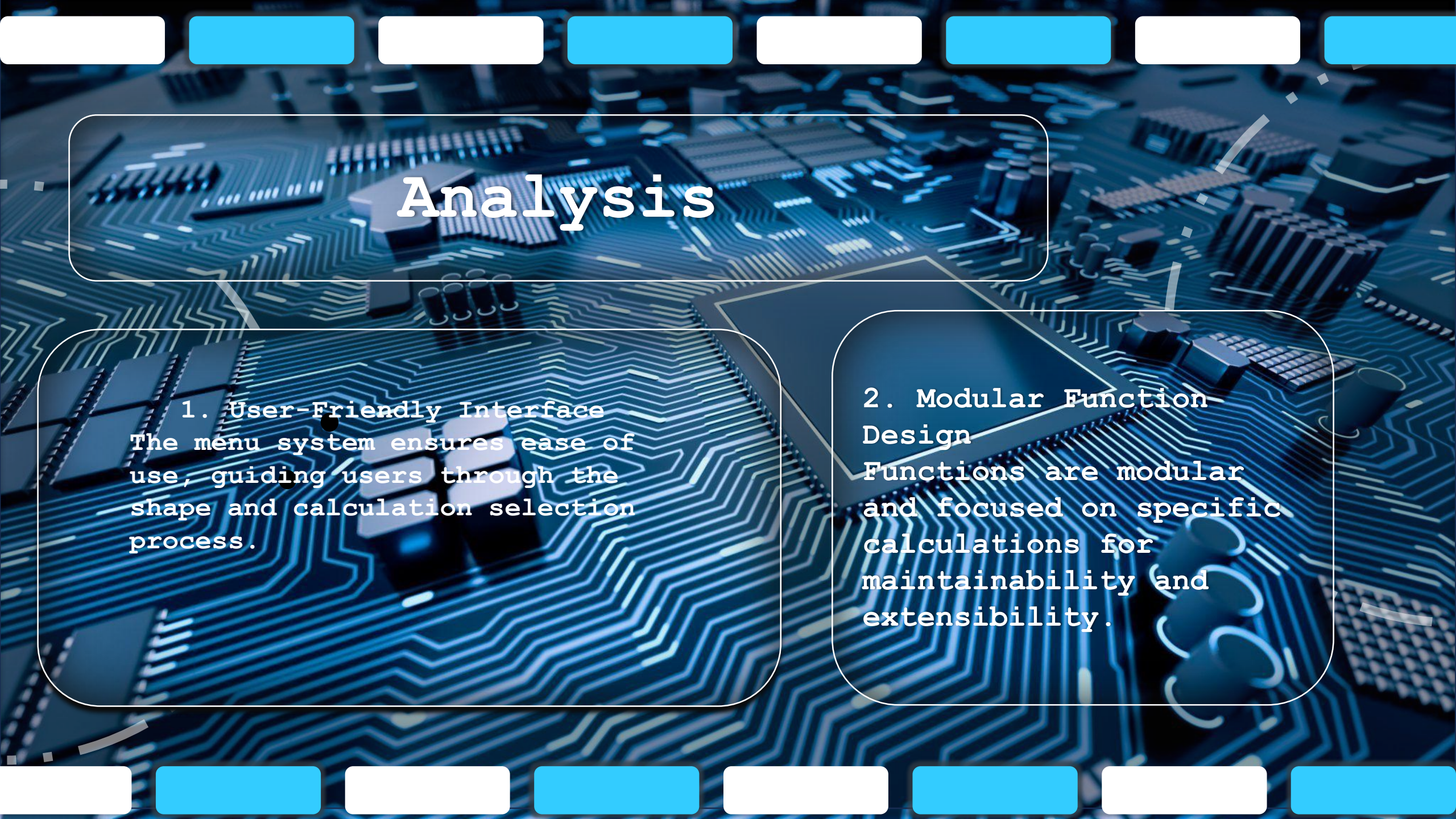
The program performs the calculation using the appropriate function.

5. Result

The calculated result is displayed to the user.

6. Continuation 6

Users are prompted if they want to continue or exit the program



Analysis

1. User-Friendly Interface

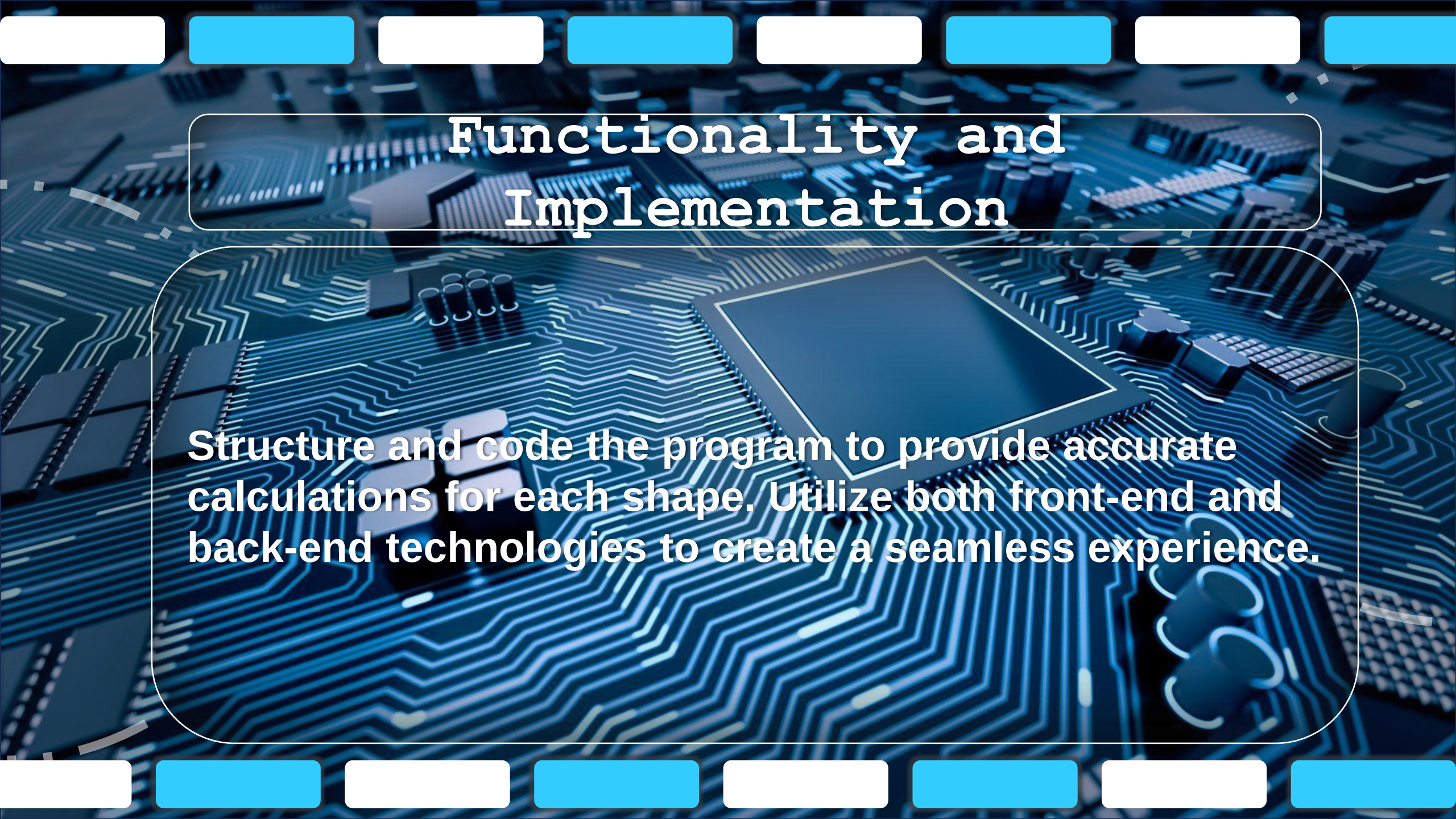
The menu system ensures ease of use, guiding users through the shape and calculation selection process.

2. Modular Function Design

Functions are modular and focused on specific calculations for maintainability and extensibility.

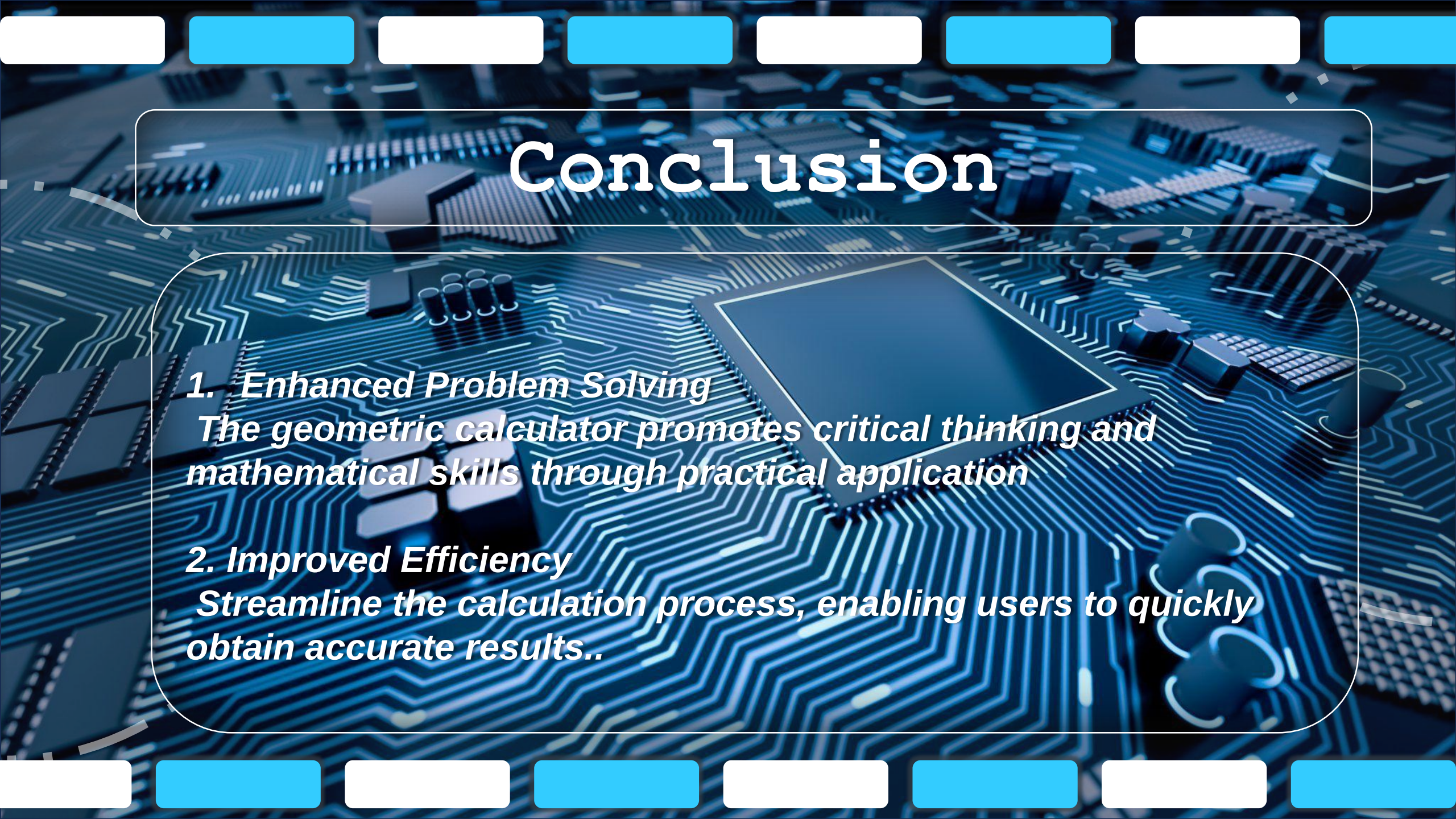
Functions to Implement

<pre>• calculateCircleArea(radius)</pre>	<pre>• calculateCircleCircumference(radius)</pre>
<pre>• calculateRectangleArea(length,width)</pre>	<pre>• calculateRectanglePerimeter(length, width)</pre>
<pre>• calculateTriangleArea(base,height)</pre>	<pre>• calculateTrianglePerimeter(side1,side2, side3)</pre>



Functionality and Implementation

Structure and code the program to provide accurate calculations for each shape. Utilize both front-end and back-end technologies to create a seamless experience.



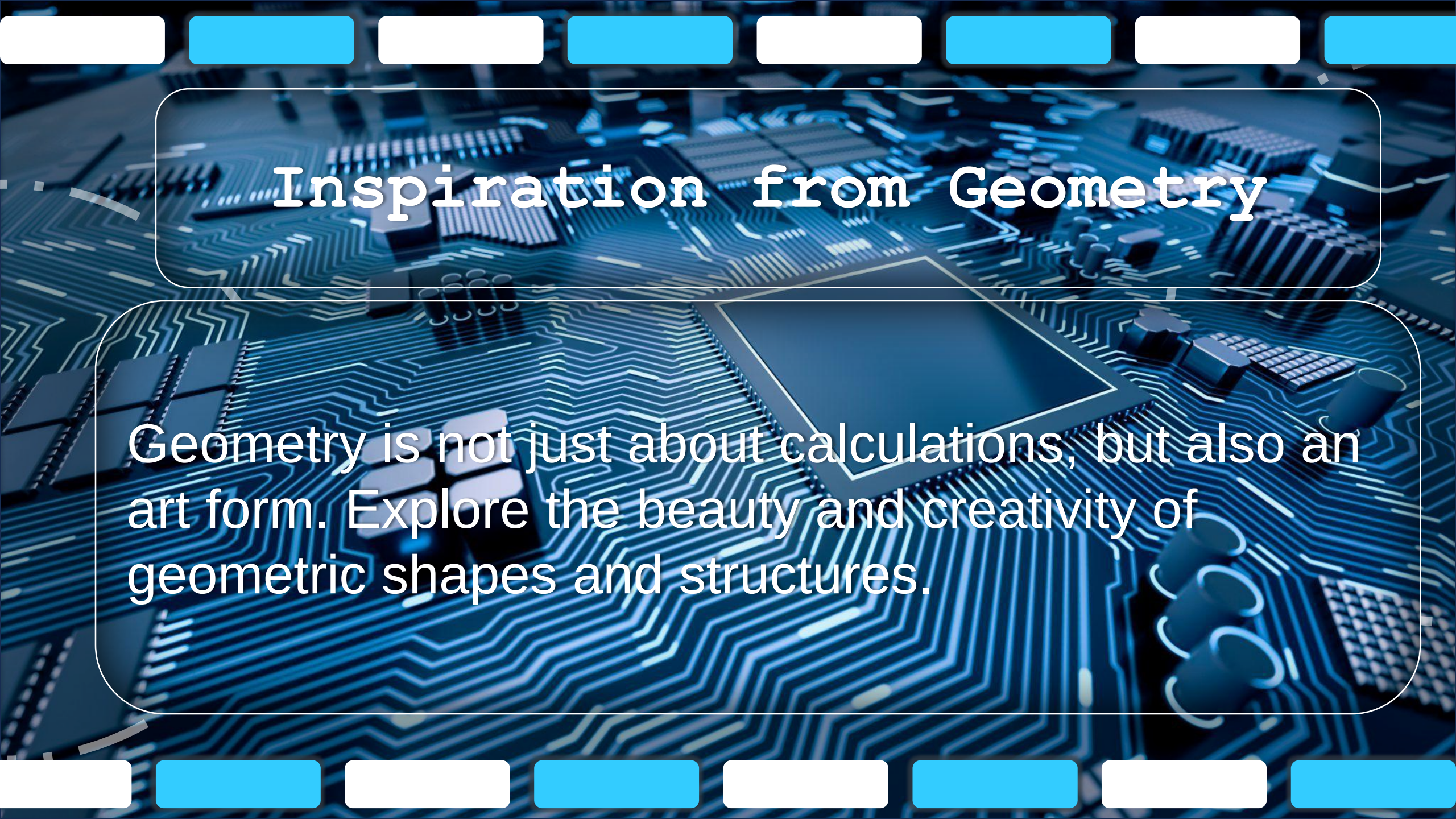
Conclusion

1. Enhanced Problem Solving

The geometric calculator promotes critical thinking and mathematical skills through practical application

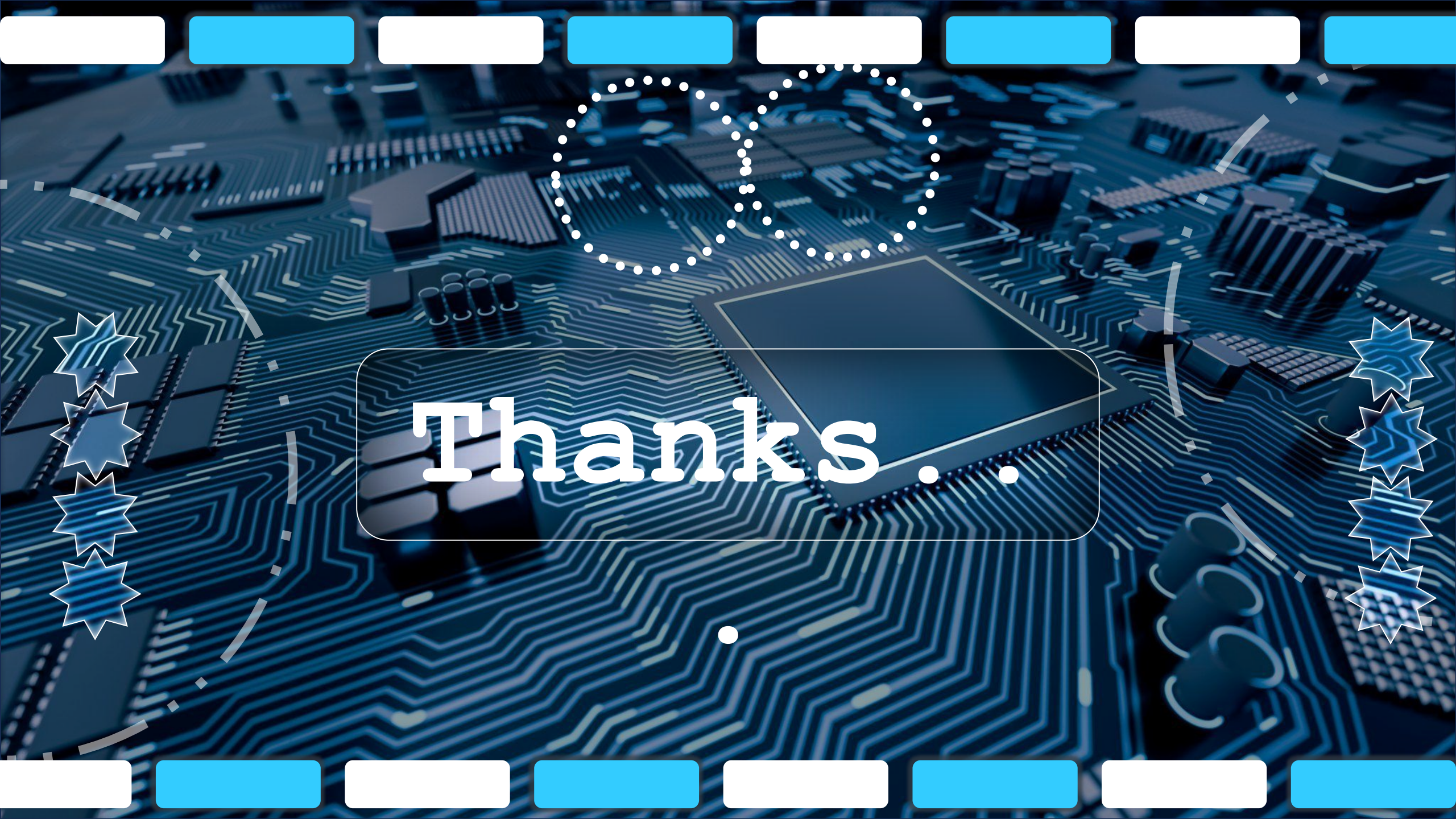
2. Improved Efficiency

Streamline the calculation process, enabling users to quickly obtain accurate results..



Inspiration from Geometry

Geometry is not just about calculations, but also an art form. Explore the beauty and creativity of geometric shapes and structures.



Thanks..